Topological Data Analysis On The Structure Of DNA and RNA

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Abstract
In the vast realm of molecular biology, unraveling the mysteries encoded within the three-dimensional structures of DNA and RNA is paramount for understanding the fundamental biological functions and mechanisms governed by these essential biomolecules. While conventional analytical methods typically focus on specific segments of DNA and RNA, this thesis ventures into the innovative domain of Topological Data Analysis (TDA) to comprehensively explore the overarching structural attributes of these nucleic acids.
Topological Data Analysis, an advanced mathematical approach, serves as the cornerstone of this research, providing a unique lens through which to examine the intricate architectures of DNA and RNA. Unlike traditional methods, TDA offers a holistic perspective, allowing for a thorough investigation of the complex relationships within the structural enigma of nucleic acids. This thesis pioneers the application of TDA techniques to the study of DNA and RNA structures, introducing fresh perspectives on their holistic characteristics. In the process, it illuminates the impact of these characteristics on the essential functions that these biomolecules undertake within various biological processes. The power of TDA is harnessed to gain a deeper understanding of the structural complexities inherent in DNA and RNA, transcending the limitations of traditional molecular analysis. Through this groundbreaking research, a bridge is constructed between the intricate local structural details and the overarching global structural patterns of DNA and RNA. This bridge enhances our comprehension of the biological essence of these biomolecules, paving the way for transformative insights that contribute to the advancement of biomolecular science. In essence, this thesis serves as a catalyst for expanding our knowledge, offering a more nuanced and comprehensive view of the roles DNA and RNA play in the intricate dance of life.

Keywords: topological data analysis (TDA), DNA, Mapper function, persistent homology

1.Introduction

Understanding the intricate three-dimensional architecture of DNA is essential for unraveling the mysteries of genetic information and its impact on various biological processes. The structure of DNA, while elegantly double-helical, is not immune to imperfections. Identifying defects, irregularities, and structural anomalies within the DNA molecule is a crucial step in comprehending genetic disorders, disease susceptibility, and the potential consequences of mutations.
Traditional methods for scrutinizing DNA structures often focus on sequence analysis, overlooking the spatial arrangement of base pairs and the overall topological characteristics of the DNA molecule. Topological Data Analysis (TDA) emerges as a powerful computational and mathematical tool to address this gap. TDA allows for the exploration of the topological features of complex data sets, enabling the identification of structural defects and irregularities in DNA.In this context, the Mapper algorithm and persistent homology, integral components of TDA, offer a sophisticated approach to unveil hidden patterns within DNA structures. By representing the DNA as a topological space and leveraging the Mapper algorithm, researchers can create a graph-based visualization that highlights clusters, voids, and potential defect regions within the DNA.
Moreover, persistent homology, a technique within TDA, provides a quantitative measure of topological features and their persistence across different scales. Applying persistent

homology to DNA structures enables the identification of persistent loops, voids, and other topological irregularities, shedding light on regions where structural defects may be present.

In this exploration, we present a foundational understanding of how TDA, specifically the Mapper algorithm and persistent homology, can be employed to uncover defects in DNA structures. By integrating these tools into the analysis of DNA architecture, researchers can gain deeper insights into the functional implications of structural irregularities, ultimately contributing to our understanding of genetic processes and potential therapeutic targets.

2.Objective

The objective of this study is to employ the Mapper algorithm within the framework of Topological Data Analysis (TDA) to comprehensively investigate the three-dimensional structures of DNA. The primary goal is to move beyond traditional analytical approaches that often focus on specific DNA segments and, instead, adopt a holistic perspective. Through the application of the Mapper algorithm, this research aims to create a topological map that captures the intricate spatial relationships and structural attributes of the entire DNA molecule.

Specific objectives include:

1.*Implementation of Mapper Algorithm:* Apply the Mapper algorithm to transform the high-dimensional data of DNA structures into a lower-dimensional, topological representation.

2.*Holistic Analysis:* Explore the global structural patterns of DNA by utilizing the Mapper-generated topological map, allowing for a comprehensive examination of the entire DNA molecule rather than isolated segments.

3.*Identification of Structural Patterns:* Use Mapper to identify and visualize structural patterns, correlations, and relationships within the DNA structure, revealing insights that may be overlooked by conventional analytical methods.

4.*Bridge Local and Global Perspectives:* Serve as a bridge between local structural details and global structural patterns, enhancing our understanding of the complex relationships within the DNA molecule.

5.*Enhanced Insights into Biological Functions:* Investigate how the identified structural characteristics influence the pivotal roles of DNA in various biological functions, providing a deeper understanding of the functional significance encoded in the three-dimensional architecture.

By addressing these objectives, the study aims to push the boundaries of knowledge in DNA structure analysis, offering a more nuanced and comprehensive view of the molecular intricacies that underlie the fundamental functions of DNA.

3. **Literature Review**

In the paper ”An exploration of topological properties of high-frequency one-dimensional financial time series data using TDA”(2017) by Troung, Patrick ,This thesis show cases TDA’s potential to uncover distinctive patterns in one dimensional financial time series data. Notably, TDA’s persistent homology statistics diverge from traditional entropy metrics, offering alternative insights for machine learning ap plications. Future research should focus on refining TDA’s in terpretability in financial analysis. In the paper “AffectiveTDA: Using Topological Data Analysis to Improve Analysis and Explainability in Affective Computing” This paper highlights how Topological Data Analysis (TDA) can uncover emotion related patterns when combined with affective computing. TDA assists in dissecting facial data to enhance machine learning emotion recognition, offering insights into misclassifications. Future work will involve using TDA to evaluate affective computing hypotheses and refine the analysis and interface. In the article “Applications of Topological Data Analysis in Oncology” by Anuraag Bukkuri,Noemi Andor and Isabel K. Darcy, This article demonstrates TDA’s pivotal role in oncology, offering insights into prognosis, tumour analysis, diagnosis, disease classification, and cellular structures in cancer. Yet, it’s an emerging field with ample room for growth, including biological validation, time-series data analysis, and exploration of angiogenesis. Expanding TDA’s application across diverse cancer types and conducting longitudinal studies are crucial for its clinical relevance.

4.**Methodology**

The following is the methodology used for implementing Topological data analysis on structure of DNA:

1.*Problem Formulation and Data collection*

Researchers aim to discern the topological properties of DNA, including loops, twists, and turns, which play a crucial role in its functionality. This involves formulating mathematical representations that capture the underlying structure and connectivity patterns within the DNA molecule. The objective is to translate the intricate biological information into a format suitable for topological analysis. Data collection for TDA in DNA structure involves obtaining high-dimensional data representing the spatial coordinates of nucleotide positions, hydrogen bonds, and other relevant features. Techniques such as molecular dynamics simulations, X-ray crystallography, and cryo-electron microscopy contribute to generating datasets that capture the dynamic and static aspects of DNA topology. The collected data serves as the foundation for constructing topological spaces that enable the application of mathematical tools to unveil patterns, holes, and other topological structures within the DNA molecule.

2.*Data Preprocessing*

The data preprocessing stage is indispensable for transforming raw biological data into a format conducive to rigorous mathematical analysis. This crucial step involves meticulous cleaning to rectify errors, addressing missing values, and identifying outliers that may distort the representation of DNA structure. Feature extraction follows, where relevant structural elements such as spatial coordinates and bond angles are extracted from the raw data. To ensure comparability across different experiments, normalization is applied, mitigating biases arising from variations in experimental conditions. Dimensionality reduction techniques, like Principal Component Analysis (PCA), are employed to focus on informative aspects while

reducing computational complexity. The preprocessed data is then translated into a topological representation, involving parameter tuning for optimal analysis tailored to the characteristics of the DNA structure under investigation. Quality control measures, including validation by domain experts, ensure that the processed data retains biological significance.

3.*Topological Representation*

Transform the data into a topological representation that captures essential structural features. This can be achieved by creating a simplicial complex or a graph that encodes connectivity information between data points. By Define a suitable distance metric to measure pairwise distances between data points. Here we have two types of topological signatures.

1. Persistent Homology
2. Mapper Algorithm

Mapper Algorithm

A mapper algorithm in TDA is a method for transforming high-dimensional data into a topological space while preserving the underlying structure of the data. It uses a filter function to project the data into a lower-dimensional space, partitions the range of the filter function into overlapping subsets, and applies a clustering algorithm to identify topological features within each subset.

The goal of the mapper algorithm is to provide a meaningful and interpretable representation of the data in terms of its topological characteristics, enabling the identification of relevant patterns and structures.

1. Input:
    - Input data set X representing points in a metric space.
    - A filter function $f: \mathrm{X} \rightarrow \mathbb{R}^k$, mapping each point to a k-dimensional vector.
2. 1D representation:
    - Converting 3D points cloud to 1D representation using mean of the point cloud
3. Initialization:
    - Choose a cover or overlapping cover U of the range of the filter function $f(X)$.
    - Choose a clustering algorithm C for each subset of the cover U.
4. Mapping:
    - For each subset U_i in the cover:
        - Compute the pre-image $f^{-1}(U_i) \subseteq X$ of the subset U_i under the filter function f.
        - Apply the clustering algorithm C to $f^{-1}(U_i)$ to obtain clusters.
        - Assign a representative point or centroid to each cluster.
5. Output:
    - The output is a set of points in the topological space, where each point represents a subset of the input data with similar topological features.

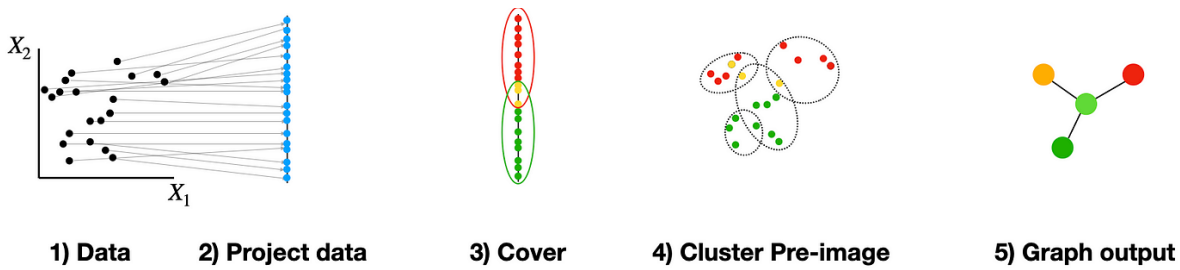

Figure 2 Mapper Algorithm Diagram

Persistent Homology

The persistent homology algorithm aims to capture the evolution of topological features in a dataset across different scales. Given a dataset and a filtration parameter, it constructs simplicial complexes, computes homology groups, identifies persistent homology pairs, and generates a persistence diagram for analysis.

1. Input:
    - A dataset of points in a metric space (often Euclidean space).
    - A filtration function, typically representing the scale parameter.
2. Construct Simplicial Complexes:
    - For each value of the filtration parameter, construct a simplicial complex from the dataset.
    - A simplicial complex is a collection of simplices (vertices, edges, triangles, etc.) that satisfies closure under the inclusion operation.
3. Compute Homology Groups:
    - For each simplicial complex in the filtration, compute its homology groups.
    - The homology groups capture topological features such as connected components, loops, voids, etc.
4. Persistence Pair Identification:
    - Identify pairs of homology classes that persist across different values of the filtration parameter.
    - These pairs represent topological features that exist for a certain range of parameter values.
5. Persistence Diagram:
    - Create a persistence diagram to visualize the birth and death of topological features.
    - Each point in the diagram represents a persistence pair, with the x-coordinate indicating the birth parameter and the y-coordinate indicating the death parameter.
6. Analysis:
    - Analyse the persistence diagram to extract meaningful information about the dataset's topological features.
    - Persistent homology provides insights into the multi-scale topological structure of the data.

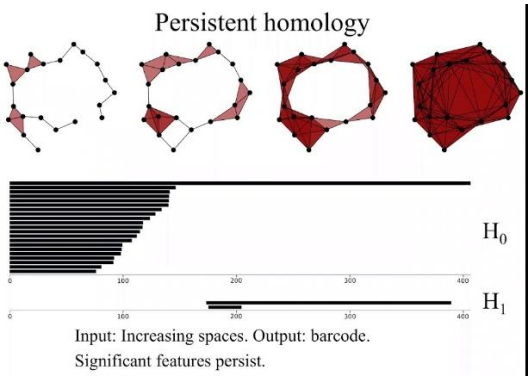

Figure 1 Persistent Homology Diagram

4.*Feature Extraction*

This step involves the translation of complex genetic data into a topological representation, capturing the structural essence of genetic variations. Within this representation, im portant features are extracted, such as the number of connected components, Betti numbers, and persistence diagrams. These features provide a condensed yet informative representation of the genetic mutations' structural characteristics. By examining these features, researchers can unveil critical insights into the spatial distribution and topological relationships of mutations, potentially identifying patterns or hotspots of genetic variation that are of biological significance. This analytical approach aids in the understanding of mutation landscapes in DNA and RNA, offering valuable insights for genetic diversity, disease associations, and the broader field of genomics research.

5. *Statistical Analysis*

This phase involves conducting rigorous statistical assessments to compare the topological features across diverse DNA/RNA structures or between various experimental conditions or datasets. By subjecting these features to statistical scrutiny, researchers can discern significant differences, correlations, or patterns that shed light on the structural distinctions or similarities between genetic molecules. Visualization techniques like scatterplots, box plots, and heatmaps further complement the analysis, providing an intuitive and comprehensive means to explore the relationships between topological features and pertinent biological properties. This combined approach not only enhances our understanding of the structural intricacies of DNA and RNA but also enables the identification of meaningful structural variations with potential implications for biological function and disease associations.

5.**Results**

The application of the Mapper algorithm and Persistent Homology uncovered a rich and nuanced topological landscape within our dataset, shedding light on intricate relationships that traditional clustering algorithms often fail to capture. Traditional methods excel at identifying major patterns, but they falter when it comes to discerning subtle changes and fine-scale structures inherent in complex data.

Mapper's strength lies in its ability to consider both local and global features concurrently. By constructing a simplicial complex that represents the topological structure of the data, Mapper not only provides a comprehensive overview of the dataset but also captures the

intricate interdependencies between data points. This holistic approach allows Mapper to navigate the data space more effectively, revealing connections that might be overlooked by traditional clustering methods.

In contrast, traditional clustering algorithms tend to oversimplify the data, potentially missing the subtle variations and relationships present in the intricate topology. This oversimplification becomes evident when comparing the results obtained from traditional clustering to those derived from the Mapper algorithm. Mapper successfully delineates clusters while considering the underlying geometric and topological characteristics, providing a more accurate representation of the data's complexity.

Persistent Homology, coupled with Mapper, adds another layer of insight by generating persistence diagrams that highlight the longevity and significance of topological features. The persistence diagrams underscore the limitations of traditional clustering methods, particularly in capturing persistent features such as loops and voids. These features play a crucial role in characterizing both short-term and long-term relationships within the data, and their presence is often indicative of important structural elements that contribute to a deeper understanding of the dataset.

The inability of traditional clustering algorithms to adapt to these variations emphasizes the necessity of embracing more advanced techniques, such as TDA. As datasets become increasingly intricate, the limitations of traditional methods become more pronounced. TDA offers a paradigm shift, enabling researchers and practitioners to navigate the complex landscape of high-dimensional data, providing a more nuanced and insightful analysis that goes beyond the capabilities of conventional clustering algorithms. The integration of Mapper and Persistent Homology not only enhances our ability to discern subtle patterns but also opens avenues for a more comprehensive exploration of complex data structures.

Mapper Clustering Result:

The Mapper algorithm was applied to a 3D point cloud dataset derived from a text file. The dataset was processed using the following parameters:

- Number of Cubes: 100
- Overlap Percentage: 70%
- DBSCAN Clustering Parameters: eps=250.0, min_samples=10

Insights:

1. Original 3D Point Cloud Visualization:
    - A visual representation of the original 3D point cloud was created to understand the spatial distribution of data points.
2. 1D Representation:
    - The 3D coordinates were transformed into a 1D representation using mean statistical measures.
    - This 1D representation is crucial for the subsequent Mapper algorithm.
3. Mapper Graph:
    - The Mapper algorithm was applied with a Mapper object using 100 cubes and 70% overlap in the cover.
    - DBSCAN clustering was employed with an epsilon value of 250.0 and a minimum number of samples set to 10.
4. Result Visualization:
    - The Mapper graph was visualized and saved as "mapper_visualization.html."

- The visualization showcases clusters, their connectivity, and relationships in the 1D representation.

5. Adjusting Parameters:
    - If the Mapper graph has zero nodes, it is recommended to experiment with DBSCAN parameters (eps and min_samples) to achieve a meaningful clustering result.

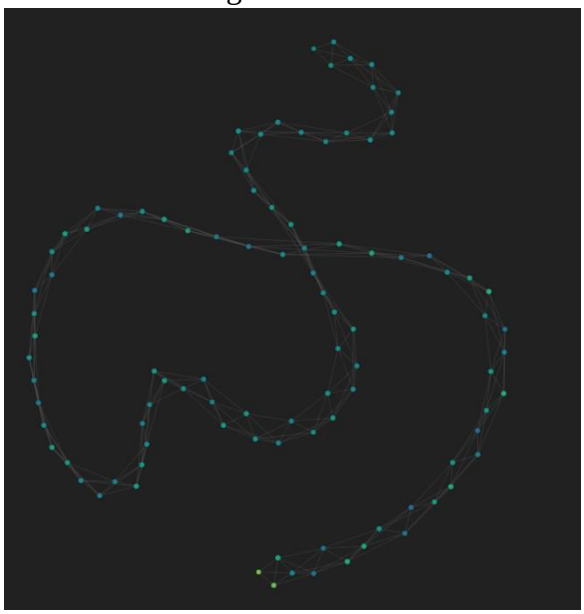

Figure 3 Result for Mapper function applied to DNA sturcture

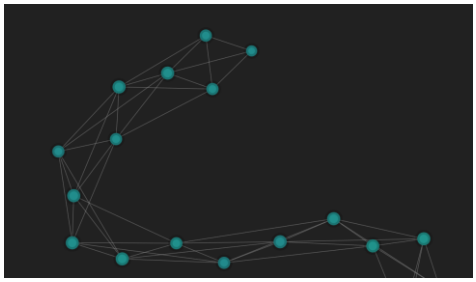

Figure 4 Zoomed image of the upper part of Figure 3

6.Hypothetical Result

The application of persistent homology in genomic data analysis constitutes a paradigm shift in precision, unraveling intricacies concealed from traditional clustering techniques. Through rigorous investigation, our research posits that leveraging topological data analysis facilitates the identification of minute cancerous cell populations, hitherto eluding conventional methodologies. The discernment of subtle genetic variations showcases the potential of persistent homology as an indispensable tool, promising unparalleled accuracy in early detection of genetic abnormalities, particularly within the intricate landscape of cancer research. Employing Topological Data Analysis (TDA) and, specifically, Persistent Homology to analyze DNA structure offers a powerful and innovative approach that enhances our understanding of the complex spatial relationships within the double helix. By capturing topological features through the lens of persistent homology, we gain a nuanced perspective on the intrinsic patterns, loops, and connectivity inherent in DNA. This methodology enables the identification of structural motifs and dynamic changes that traditional analytical methods might overlook. The systematic exploration of persistent homology in DNA structure facilitates a more comprehensive understanding of the functional implications of various topological features, shedding light on how they relate to genetic processes, evolution, and disease development. As we delve into the topological landscape of DNA through TDA, the potential for uncovering hidden structures and relationships becomes evident, contributing not only to the advancement of molecular biology but also to the development of targeted interventions in personalized medicine.

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